

A genome-wide, guide-site-aware CRISPRa targetability atlas across murine microglial ATAC-seq contexts

Ernest Darell Zermeño^{1,2,*}

¹Universidad Panamericana, Guadalajara, Mexico

²Centro Universitario de Ciencias Biológicas y Agropecuarias, Universidad de Guadalajara,
Zapopan, Mexico

*Correspondence: 0244552@up.edu.mx; ORCID: 0009-0002-1721-4526

Abstract

Background: Compact CRISPR-mediated transcriptional activation (CRISPRa) systems can ease adeno-associated virus packaging constraints, but their restricted protospacer-adjacent motifs (PAMs) and the chromatin environment of the actual guide site both affect candidate selection. Existing design tools are general purpose; a reproducible resource focused on microglial regulatory contexts has been lacking.

Results: We uniformly reprocessed 13 public murine microglial ATAC-seq runs from three studies and scanned the transcription-oriented -400 to -50 bp promoter window of 21,599 GENCODE vM33 protein-coding genes. The atlas represents five distinct nuclease/PAM classes across six modelled CRISPRa configurations; HEAL and SminiCRa share the Un1Cas12f1 TTTR targeting class, and the dNme2Cas9 activator is a sequence-level proposed configuration. Unlike a promoter-midpoint proxy, the primary definition requires the complete protospacer plus PAM interval to be contained in a primary ATAC peak. In a frozen 55-gene therapeutic panel, 50/55 genes (90.9%) contained a sequence-filtered TTTR candidate, whereas 41.8–65.5% had a complete Un1Cas12f1 guide site supported across individual dataset contexts. The corresponding genome-wide range was 29.0–43.3%. *Tfeb* and *Tfe3* had TTTR candidates but primary peak support in 0/6 and 6/6 contexts, respectively. Canonical-TSS, APPRIS, legacy-TSS, matched-depth, and Genrich/MACS3 analyses showed explicitly which calls were definition-sensitive.

Across-study differences remained descriptive because study, isolation, sequencing, depth, and biology were confounded.

Conclusions: PAM availability was broader than guide-site support in the promoter window studied. The atlas provides auditable candidate protospacers and robustness annotations for experimental prioritization, not evidence of CRISPRa efficacy, non-activatability, safety, or therapeutic benefit.

Keywords: CRISPRa; microglia; ATAC-seq; chromatin accessibility; guide design; Cas12f; PAM; targetability atlas

1 Background

CRISPR-mediated transcriptional activation (CRISPRa) increases endogenous gene expression without introducing a targeted double-strand break. Its use in vivo is constrained by delivery: the coding sequence of SpCas9 alone occupies much of the approximately 4.7-kb adeno-associated virus payload, motivating smaller effectors and activation architectures [1, 2]. Compact systems based on Un1Cas12f1 and CjCas9 have demonstrated mammalian transcriptional activation, including HEAL, SminiCRa, and MiniCAFE [3–5]. Compactness does not remove the need for an appropriately positioned PAM and protospacer.

Chromatin provides a second selection layer. CRISPRi/a activity depends strongly on guide position and local regulatory context, and established design frameworks already use chromatin data to prioritize guide RNAs [6–8]. Accessibility is nevertheless probabilistic rather than a universal binary requirement: an ATAC-supported site is a more plausible candidate, while absence of an ATAC peak does not prove that a potent activator cannot engage the locus.

Microglia are especially relevant because their regulatory landscape responds to tissue environment and inflammatory history [9–11]. CRISPRi/a has also been implemented in human induced-pluripotent-stem-cell-derived microglia, establishing a practical route to functional follow-up [12]. Public murine ATAC-seq datasets span homeostatic, endotoxin-exposure, surgical-control, and stroke settings, but these datasets were generated by different laboratories with different isolation and sequencing strategies. They can support a multi-context resource, not a balanced causal comparison of “microglial states.”

Here we rebuild a genome-wide murine microglial atlas around the coordinate that matters experimentally: the complete protospacer–PAM interval. We distinguish five targeting classes across six modelled CRISPRa configurations, define TSS and peak-set provenance, use independent biological replicates where available, label technical-only contexts, and quantify sensitivity to depth, peak caller, TSS choice, and operational definition. The contribution is a transparent resource for prioritizing experiments rather than a new guide-scoring algorithm or a claim of functional activation.

2 Results

2.1 The revised atlas makes guide position, TSS choice, and replication explicit

The workflow starts from checksum-verified FASTQs and GENCODE vM33 rather than inherited peak files (Figure 1). It includes 21,599 nuclear protein-coding genes, 13 sequencing runs, six dataset contexts, and a frozen 55-gene panel. Ensembl-canonical transcripts define the primary TSS. APPRIS selected a different TSS for 2631 genes and the previous most-5-prime GENCODE-basic rule differed for 4596 genes, demonstrating that TSS selection is not a bookkeeping detail.

Five nuclease/PAM classes are evaluated (Table 1). HEAL and SminiCRa both use Un1Cas12f1 with a TTTR PAM, so their sequence and ATAC-based targeting matrices are identical even though their scaffolds and activation architectures differ. They are consequently represented as two configurations but one targeting class. Nme2Cas9 contributes a compact, experimentally established N₄CC DNA-targeting rule, but the dNme2Cas9 activator entry is a proposed configuration rather than a validated CRISPRa platform [13].

For biologically replicated inputs, the primary peak set retains sequence supported by a qualifying pair of independent replicate peaks at 50% reciprocal overlap. The intersection of each qualifying pair makes complete guide containment conservative. The PBS/PBS, PBS/LPS, and LPS/LPS inputs consist of paired technical runs from a single biological preparation per condition; they are pooled and explicitly carry no biological-reproducibility claim.

2.2 Sequence-level PAM coverage is broad in the surveyed promoter window

Both strands of the -400 to -50 bp window were scanned with IUPAC-aware PAM matching and class-specific spacer lengths. Candidates passed prespecified GC, homopolymer, and cloning-site filters. Across the genome, 20022 of 21,599 promoters (92.7%) contained at least one passing Un1Cas12f1 TTTR candidate (Figure 2A). Coverage of the frozen panel was 50/55 (90.9%; Figure 2B); the corresponding values were 96.4%, 100.0%, 100.0%, and 100.0% for SaCas9, SpCas9, CjCas9, and Nme2Cas9. These values quantify sequence availability within the chosen window; they do not predict expression gain.

Candidate abundance varied substantially among PAM classes (Figure 2D). This was expected from motif degeneracy and spacer filtering, and it is why the main comparison is not simply whether a promoter contains a motif. The relevant atlas question is whether a specific complete candidate interval has reproducible ATAC support.

2.3 Complete guide-site support is narrower than PAM availability

The primary call requires at least one complete protospacer plus PAM interval to be fully contained in the applicable primary peak. For Un1Cas12f1, 41.8–65.5% of the frozen panel met this rule across the six individual contexts, compared with 90.9% at the sequence-only layer (Figure 3A). Genome-wide Un1Cas12f1 guide-site support ranged from 29.0% to 43.3% (Figure 2C).

Within a context, the range among the five nuclease/PAM classes was 10.9–14.5 percentage points in the panel. Across contexts, the Un1Cas12f1 range was 23.6 percentage points. This does *not* establish that biological state has a larger effect than nuclease choice: the cross-context range also contains laboratory, sorting, sequencing-layout, depth, and population-composition differences. Accordingly, Figures 2 and 3 label the six columns by study and treatment rather than presenting them as exchangeable states.

The guide-by-context heatmap gives the intended use of the resource (Figure 3C). A positive cell means that at least one candidate interval satisfies the primary rule in that context. A negative cell means only that no candidate passed this particular sequence, window, TSS, peak, and containment definition.

2.4 TSS, containment, depth, and peak caller expose definition-sensitive loci

Four operational quantities were kept separate: complete guide containment, any guide/peak overlap, promoter-midpoint coverage, and any promoter/peak overlap (Supplementary Figure S3). They answer different questions and produced different estimates; promoter overlap was not relabelled as guide targetability.

TSS choice had a locus-specific impact. The Ensembl-canonical and legacy TSSs for *Tfeb* differed by 48690 bp, whereas the corresponding *Tfe3* definitions differed by 0 bp (Figure 4C). Under the canonical primary analysis, *Tfeb* contained 5 passing TTTR candidates and had guide-site support in 0/6 contexts; *Tfe3* contained 3 and had support in 6/6. These are target-selection hypotheses, not evidence that either paralog is functionally superior. The distinction matters because TFEB and related MiT/TFE factors regulate lysosomal programs, but abundance, localization, occupancy, and downstream function cannot be inferred from promoter ATAC alone [14–16].

To evaluate sequencing depth and caller dependence without discarding replication structure, biological inputs were downsampled at replicate level to an equal condition-total target, called per replicate, and reconstructed with the same replicate-consensus rule. Technical-only contexts were downsampled after cross-lane pooling and duplicate removal. Genrich and MACS3 were then evaluated in parallel. Across the TSS, depth, and caller variants, the largest change relative to a primary panel call affected 16 genes in a class-context cell, corresponding to 29.1 percentage points (Supplementary Figure S4). The Jaccard index of promoters overlapped by each caller and binary promoter-call concordance are reported separately (Supplementary Figure S6), rather than justifying Genrich from a local software failure.

2.5 Panel patterns and candidate rankings remain descriptive

For Un1Cas12f1, 19 panel genes had guide-site support in all six contexts, 15 had support in at least two studies without being positive in every context, 5 were supported within one study, and 16 had no support under the surveyed primary definitions (Figure 4A). These labels replace causal categories such as “inflammation-gained” or “injury-open.” A within-panel Fisher analysis evaluated whether these support patterns were associated with the functional labels used to curate the panel, with Benjamini–Hochberg adjustment across the full test family (Supplementary Figure S5).

134 It was treated as descriptive because the panel and its categories were selected a priori.

135 Candidate Table S3 reports the selected transcript, TSS, spacer and PAM coordinates, strand,
136 distance to TSS, supporting peak identifier and signal in each context, distance to the peak summit,
137 and the number of positive contexts. A normalized candidate-by-run companion table reports
138 complete and partial support, run-level peak identifier, signal, and summit distance, preserving
139 the distinction between biological replicates and technical lanes. The first five candidates per gene
140 and targeting class are presented as candidate protospacers rather than recommendations. The
141 Un1Cas12f1 subset additionally received exhaustive Bowtie alignment through three mismatches
142 followed by an orientation-aware TTTR check and exon/canonical-promoter annotation (Figure 5).
143 That screen does not model bulges, cleavage, nuclease-specific mismatch energetics, chromatin at
144 off-targets, or experimental genotoxicity.

145 2.6 Matched-promoter analysis separates panel composition from random ge- 146 nomic placement

147 The previous comparison with peaks shuffled uniformly across the genome was replaced because
148 open-chromatin peaks are naturally promoter enriched. For each frozen-panel promoter, non-panel
149 promoters were matched using promoter GC decile, annotated transcript-count bin, and number of
150 passing Un1Cas12f1 candidates. Ten thousand matched panels were sampled with seed 1729 (Sup-
151 plementary Figure S7). The analysis compares this curated panel with sequence- and annotation-
152 matched promoters; it is not functional validation and could not additionally match expression or
153 mappability across every context.

154 3 Discussion

155 This revision changes the scientific object from a promoter-level proxy to a guide-site resource.
156 The old midpoint rule could reject a fully accessible candidate because a peak failed to cover -225
157 bp, or accept promoter accessibility without proving that any complete candidate was supported.
158 Requiring the complete spacer-PAM interval removes that mismatch while preserving midpoint and
159 overlap quantities as transparent sensitivities.

160 The broad sequence coverage of compact PAM classes is useful but unsurprising in a 350-bp, two-

strand search. The more informative output is the identity and robustness of individual sites after integrating public microglial ATAC-seq. ATAC support is not a universal binary determinant. Nucleosome breathing, activator potency, chromatin remodelers, and experimental context can permit activity where a peak caller reports no peak. Conversely, an accessible site can fail because of guide folding, effector geometry, delivery, expression, or biological feedback. “Targetable” is therefore an operational atlas label, and candidate-site plausibility is the more general interpretation.

The multi-context design has a deliberate limit. The three studies were not a joint experiment. Gosselin used ex vivo homeostatic microglia and single-end sequencing; Zhang X/Kracht assessed endotoxin exposure with technical sequencing runs; Zhang and colleagues profiled separate sham and stroke animals using a different isolation strategy. Downsampling, a shared peak universe, and a second caller diagnose some technical sensitivity but cannot remove study confounding. The strongest comparisons are those internal to the Zhang X/Kracht and sham–stroke series, while consistency across laboratories is useful as robustness evidence rather than a causal state effect.

The revised *Tfeb/Tfe3* example illustrates why the additional safeguards matter. A large alternative-TSS displacement can move the entire 350-bp search window to another locus segment. Primary support for one paralog across more contexts would still not demonstrate greater activation, lysosomal flux, or therapeutic benefit. The atlas instead defines a direct experiment: select candidates with explicit TSS and peak support, activate endogenous *Tfeb* and *Tfe3* side by side in an appropriate microglial system, and measure transcriptional engagement and function.

Several limitations remain. The promoter window is standardized rather than transcript- and cell-state-optimized. Canonical/APPRIS annotations are proxies for the expressed TSS, because matched microglial CAGE data were unavailable across all contexts. Replicate-consensus peaks increase specificity at the cost of sensitivity. Three contexts lack biological replicates. ATAC signal is relative and peak-caller dependent. The 55-gene panel is curated rather than randomly sampled, so its resampling intervals measure stability of that fixed panel rather than population inference. The promoter-matched null lacks expression and mappability matching. Candidate scores are heuristic for Un1Cas12f1, and off-target analysis remains preliminary. Finally, all results are murine and computational; direct CRISPRa and functional validation are required.

4 Conclusions

Within a standardized murine promoter window, PAM-bearing candidates were common, whereas complete guide sites supported by primary microglial ATAC peaks were less frequent and definition dependent. The revised atlas makes those dependencies visible at guide resolution and provides exact candidates, peak provenance, TSS alternatives, quality controls, and caller/depth sensitivities. It should be used to prioritize experiments, not to infer efficacy or rule out loci.

5 Methods

5.1 Study design and frozen panel

The analysis was designed as a predictive genomics resource. A pre-existing 55-gene panel covering autophagy/lysosome, inflammation, phagocytosis, lipid metabolism, neuroprotection, microglial identity, and epigenetic regulation was frozen in `config/therapeutic_genes_locked.csv` before the present reanalysis. The panel is a complete curated set of interest, not a random sample of therapeutic genes. Genome-wide results are therefore reported alongside panel summaries.

5.2 Reference genome, transcript selection, and promoter windows

The workflow downloads GRCm39/mm39 from UCSC and GENCODE mouse vM33 annotations [17]. Nuclear protein-coding transcripts were grouped by gene symbol. The primary transcript was tagged `Ensembl_canonical`; genes without that tag used APPRIS principal, then a GENCODE-basic fallback with transcript-support level and length as deterministic tie breakers. Sensitivities independently selected APPRIS principal and the most 5-prime GENCODE-basic transcript. Every selection, fallback, transcript identifier, strand, TSS, tag set, and transcript count is recorded in `reference/tss_selection.tsv`.

TSSs use zero-based genomic coordinates. The promoter search interval was -400 to -50 bp in transcriptional orientation: $[TSS-400, TSS-50)$ on the positive strand and the corresponding downstream genomic coordinates on the negative strand.

5.3 Nuclease/PAM classes and candidate scanning

Five distinct targeting classes were scanned (Table 1). PAM and spacer rules followed the corresponding experimental literature [4, 5, 13, 18]. Overlapping PAM matches were detected on both strands using full IUPAC reverse complements. Class-specific spacers were extracted only when the full spacer and PAM fell within the promoter window. Protospacers were retained at 30–70% GC, with no homopolymer longer than four nucleotides and no BsmBI or BsaI recognition site in either orientation. Candidate counts therefore refer to sequence-filtered protospacers, not raw PAM occurrences. Only Un1Cas12f1 candidates received a simple heuristic score based on GC balance, homopolymer length, and poly-T avoidance; no validated efficacy model was claimed.

5.4 Public ATAC-seq inputs

Thirteen runs were obtained from ENA using URLs and MD5 checksums frozen in `config/samples.tsv` (Table 2). The Gosselin dataset comprised two independent single-end homeostatic microglial runs (GSE89960) [9]. The Zhang X/Kracht dataset comprised paired technical runs for PBS/PBS, PBS/LPS, and LPS/LPS preparations (GSE175578) [11]. The Zhang dataset comprised two sham and three stroke biological replicates (GSE220041) [19].

5.5 ATAC-seq processing and quality control

ATAC-seq processing followed established principles [20, 21]. Adapters were removed with Trim Galore 0.6.10 [22]. Reads were aligned with Bowtie2 2.5.4 in `--very-sensitive` mode; paired-end data additionally used `--maxins 1000 --no-mixed --no-discordant` [23]. BAMs were sorted with SAMtools 1.21, duplicate fragments removed with Picard 3.3, and alignments filtered to $\text{MAPQ} \geq 30$, proper pairs for paired-end data, primary/non-supplementary alignments, and non-mitochondrial sequence [24, 25]. Optical-duplicate inference was disabled because ENA/SRA read identifiers do not retain flowcell coordinates; coordinate-based duplicate removal was unchanged. Paired-end blacklist filtering removed the intact pair when either mate overlapped a blacklist interval. The official ENCODE mm10 v2 blacklist (ENCFF543DDX; SHA-256 9638bbeb...bfac7) was mapped with the UCSC mm10-to-mm39 chain: 3,360 of 3,435 intervals mapped and 75 did not; all source and mapped coordinates are recorded in `reference/blacklist_liftover_provenance.tsv`

[26]. Genrich 0.6.1 was run in ATAC mode, which applies its ATAC cut-site adjustment internally; single-end data additionally used `-y` [27]. Genrich was selected as the prespecified primary caller because it directly supports ATAC-seq, not because another caller failed locally.

Per-run QC comprised Bowtie2 overall alignment rate, usable fragment/read count, duplication metrics, replicate peak count, fraction of usable fragments or reads overlapping the run’s peaks, fragment-length summaries for paired-end data, and maximum TSS enrichment from Tn5-adjusted insertion sites within ± 2 kb. FRIIP reference lines are descriptive because library type and peak definition differ among datasets.

5.6 Primary peaks and continuous shared-universe signal

For Gosselin, sham, and stroke, peaks from distinct biological replicates were paired when reciprocal overlap was at least 0.50. Each qualifying pair contributed its intersected interval; overlapping intersections were merged without bridging unsupported gaps. Every retained base was therefore covered by at least one qualifying pair. For stroke, support in any two of three independent replicates was sufficient. For each technical-only Zhang X/Kracht preparation, filtered lanes were merged and deduplicated again at the pooled-library level so that a fragment duplicated across lanes was retained only once. Each technical pool was then called once, with an explicit absence of biological replication. This second deduplication was not applied across independent biological replicates.

Condition-level filtered BAMs were also merged to create CPM-normalized 10-bp bigWigs for visualization. A union of the six primary peak sets was merged into a shared peak universe, and read/fragment signal was quantified for every condition. This continuous layer supports inspection of peak-threshold effects but was not substituted for biological replication.

5.7 Primary targetability and operational sensitivities

For every gene, targeting class, and context, a candidate was primarily supported when its complete half-open genomic interval spanning the protospacer and PAM was contained within a primary peak. A gene-level primary call was positive when at least one candidate was supported. Separately stored sensitivities were: any partial candidate/peak overlap, promoter-midpoint coverage, and any promoter-window/peak overlap. The latter two are promoter-accessibility measures, not guide targetability.

Guide-level outputs include exact mm39 coordinates, strand, PAM, protospacer, TSS distance, transcript and TSS provenance, peak identifier, peak signal, overlap, and summit distance at both primary-context and individual-run levels. Up to five candidates per gene/class were ranked by the number of primary positive contexts, partial-overlap contexts, minimum available supporting signal, Un1Cas12f1 heuristic score where applicable, GC balance, preferred TSS distance, and coordinate. The ranking is predictive and does not constitute an experimental recommendation.

5.8 Matched-depth and peak-caller sensitivities

Filtered condition BAMs were counted as read units for single-end data and read-pair units for paired-end data. The minimum condition total was the common target. For biologically replicated contexts, this target was allocated as evenly as available depth permitted among the independent runs; each run was deterministically subsampled, called separately, and reduced to a two-replicate, 50%-reciprocal-overlap consensus. Technical-only contexts were subsampled once after cross-lane pooling and duplicate removal. Requested and observed post-sampling totals and all deterministic seeds are reported. Peaks were called independently with Genrich and MACS3 3.0.2 [28]. MACS3 used paired-end mode for paired libraries and a fixed shift/extension model for single-end libraries. The atlas was rebuilt for both matched-depth peak sets. Caller concordance was summarized by the Jaccard index of promoter sets with any caller peak and by binary promoter-window concordance; no peak-interval Jaccard was claimed.

5.9 Statistical summaries and robustness

For each class-context cell, 10,000 percentile resamples of the 55-gene vector were generated with seed 1729. These are labelled fixed-panel resampling stability intervals and are not confidence intervals for a population of all therapeutic genes. PAM-bearing but unsupported counts were reported descriptively because primary guide support is nested within sequence availability.

For the matched-promoter null, promoter GC was calculated from mm39 and combined with the number of annotated transcripts and number of passing Un1Cas12f1 candidates. Non-panel genes were matched by GC decile, transcript-count bin, and protospacer-count bin, with nearest-neighbour fallback in standardized feature space when a stratum was sparse. Ten thousand non-replacement panels were drawn where possible. Empirical p values used $(b + 1)/(N + 1)$. Expression and

mappability were unavailable on a uniform gene-by-context basis and were not matching variables.

Functional-category associations used two-sided Fisher exact tests within the frozen 55-gene universe, with Benjamini–Hochberg correction across every pattern–category combination. TSS, matched-depth Genrich, and matched-depth MACS3 calls were compared directly with the canonical primary matrix to report changed genes and proportion differences.

5.10 Preliminary off-target screen

Up to five reported Un1Cas12f1 candidates per panel gene were aligned exhaustively to mm39 with Bowtie 1.3.1 using up to three mismatches [29]. Alignments were retained only when a correctly oriented TTTR PAM was present next to the aligned protospacer. The intended coordinate was removed, and remaining alignments were annotated for overlap with a GENCODE exon or a ± 1 -kb canonical-TSS window. The screen does not evaluate bulges, chromatin, cleavage probability, or nuclease-specific safety and should be followed by purpose-built tools and experimental assays.

5.11 Reproducibility and use of generative AI

The full workflow is implemented in Snakemake 8.25 [30]. Software versions are pinned in `environment.yml`; accessions, MD5 checksums, random seeds, TSS choices, matched-depth realizations, caller concordance, shared-universe signal, source tables, and figure scripts are version controlled. Core regression tests cover promoter orientation, complete guide containment, IUPAC reverse complementation, and replicate-consensus behavior. The versioned code release and machine-readable output tables are identified in Data availability.

During revision, OpenAI ChatGPT and Codex were used for language editing, literature discovery, code drafting and refactoring, test design, and workflow documentation. The author inspected the primary sources, reviewed the code and outputs, corrected errors, and takes full responsibility for the manuscript and analysis. No generative-AI system is an author.

6 Availability of data and materials

Raw ATAC-seq data are publicly available under GSE89960, GSE175578, and GSE220041; exact ENA run URLs and MD5 checksums are in `config/samples.tsv`. The GRCm39 FASTA is down-

loaded from UCSC and GENCODE vM33 from GENCODE. Source code, frozen configuration,
source tables, and figure code are archived in the versioned release v2.0.0 at <https://github.com/Jefemaestro33/Paper0-CRISPRa-Targetability-Atlas/releases/tag/v2.0.0>. Raw FASTQs
and derived BAMs/bigWigs are not stored in Git but are recreated by workflow/Snakefile.

Declarations

Ethics approval and consent to participate

No new animal or human participants, specimens, or interventions were used. This secondary
computational analysis used public, de-identified sequencing data generated under the approvals
reported by the original studies.

Consent for publication

Not applicable.

Competing interests

The author declares no competing interests.

Funding

This research received no external funding.

Authors' contributions

E.D.Z. conceived the study, curated the target panel, designed and implemented the workflow,
performed the analysis, interpreted the results, and wrote and revised the manuscript.

Acknowledgements

The author thanks the investigators who generated and publicly released the ATAC-seq datasets
used here.

Abbreviations

ATAC-seq, assay for transposase-accessible chromatin using sequencing; CRISPRa, CRISPR-mediated transcriptional activation; FRiP, fraction of reads in peaks; PAM, protospacer-adjacent motif; TSS, transcription start site.

Additional files

Table S1. Frozen 55-gene panel, functional metadata, and descriptive guide-site support pattern.

Table S2. Complete genome-wide gene $\times$ targeting-class $\times$ context atlas, including sequence counts and every operational accessibility definition.

Table S3. Candidate protospacers with exact TSS, interval, peak, context, and preliminary off-target annotations; accompanying run-level peak-evidence and full PAM-valid alignment tables.

Table S4. Per-run ATAC-seq quality control and primary-peak provenance.

Table S5. Descriptive cross-context Un1Cas12f1 support patterns for the frozen panel.

Table S6. Original mm10 and lifted mm39 ENCODE blacklist intervals.

Table S7. Fixed-panel stability, PAM-to-guide-site attrition, matched-promoter null, and sensitivity records.

References

- [1] Suhani Vora, Jenny Cheng, Ru Xiao, Nathan J VanDusen, Luis Quintino, William T Pu, Luk H Vandenberghe, Alejandro Chavez, and George M Church. Rational design of a compact CRISPR-Cas9 activator for AAV-mediated delivery. *bioRxiv*, 2018. doi: 10.1101/298620.
- [2] Simon Böhm, Valentina Splith, Lisa M Riedmayr, Stylianos Michalakis, et al. A gene therapy for inherited blindness using dcas9-vpr-mediated transcriptional activation. *Science Advances*, 6(34):eaba5614, 2020. doi: 10.1126/sciadv.aba5614.

- [3] Hang Wan, Deqiang Kong, Tao Yan, Yang Zhou, et al. A compact and inducible dCas12f-based CRISPRa platform for programmable in vivo gene activation. *Nature Communications*, 17:1447, 2026. doi: 10.1038/s41467-025-68183-5.
- [4] Xiangnan Wang, Lingyun Li, Li Guo, Ying Feng, Zengming Du, Wei Jiang, Xia Wu, et al. Robust miniature Cas-based transcriptional modulation by engineering Un1Cas12f1 and tethering Sso7d. *Molecular Therapy*, 32(4):910–919, 2024. doi: 10.1016/j.ymthe.2024.02.013.
- [5] Shuqin Zhang et al. MiniCAFE, a CRISPR/Cas9-based compact and potent transcriptional activator, elicits gene expression in vivo. *Nucleic Acids Research*, 49(7):4171–4185, 2021. doi: 10.1093/nar/gkab174.
- [6] Max A Horlbeck, Luke A Gilbert, Jacqueline E Villalta, Britt Adamson, Ryan A Pak, Yanxia Chen, Alexander P Fields, Chong Yon Park, Jacob E Corn, Martin Kampmann, and Jonathan S Weissman. Compact and highly active next-generation libraries for CRISPR-mediated gene repression and activation. *eLife*, 5:e19760, 2016. doi: 10.7554/eLife.19760.
- [7] Kendall R Sanson, Ruth E Hanna, Mudra Hegde, Katherine F Donber, Christine Strand, Megan E Sullender, Emma W Vaimberg, Amy Goodale, David E Root, Federica Piccioni, and John G Doench. Optimized libraries for CRISPR-Cas9 genetic screens with multiple modalities. *Nature Communications*, 9:5416, 2018. doi: 10.1038/s41467-018-07901-8.
- [8] Luke Hoberecht, Pirunthan Perampalam, Aaron Lun, and Jean-Philippe Fortin. The crispr-Verse: a comprehensive Bioconductor ecosystem for the design of CRISPR guide RNAs across nucleases and technologies. *Nature Communications*, 13:6568, 2022. doi: 10.1038/s41467-022-34320-7.
- [9] David Gosselin, Dylan Skola, Nicole G Coufal, Inge R Holtman, Johannes CM Schlachetzki, Eniko Sajti, Baptiste N Jaeger, Carolyn O’Connor, Conor Fitzpatrick, Monique P Pasillas, et al. An environment-dependent transcriptional network specifies human microglia identity. *Science*, 356(6344):eaal3222, 2017. doi: 10.1126/science.aal3222.
- [10] Ann-Christin Wendeln, Karola Degenhardt, Laura Kauber, Michael Gerber, et al. Innate

immune memory in the brain shapes neurological disease hallmarks. *Nature*, 556(7701):332–338, 2018. doi: 10.1038/s41586-018-0023-4.

[11] Xiaoming Zhang, Laura Kracht, Antonio M Lerario, Marissa L Dubbelaar, et al. Epigenetic regulation of innate immune memory in microglia. *Journal of Neuroinflammation*, 19:111, 2022. doi: 10.1186/s12974-022-02463-5. ATAC-seq of FACS-purified microglia from LPS tolerance model. GEO: GSE175578.

[12] Nina M Dräger, Sydney M Sattler, Cindy Tzu-Ling Huang, Olivia M Teter, Kun Leng, Sayed Hadi Hashemi, Jason Hong, Giovanni Aviles, Claire D Clelland, Lihong Zhan, et al. A crispr/a platform in human ipsc-derived microglia uncovers regulators of disease states. *Nature Neuroscience*, 25:1149–1162, 2022. doi: 10.1038/s41593-022-01131-4.

[13] Alireza Edraki, Aamir Mir, Raed Ibraheim, Ildar Gainetdinov, Yeonsoo Yoon, Chun-Qing Song, Yueying Cao, Judith Gallant, Wen Xue, Jaime A Rivera-Pérez, and Erik J Sontheimer. A compact, high-accuracy Cas9 with a dinucleotide PAM for in vivo genome editing. *Molecular Cell*, 73(4):714–726.e4, 2019. doi: 10.1016/j.molcel.2018.12.003.

[14] Marco Sardiello, Michela Palmieri, Alberto Di Ronza, Diego L Medina, Marta Valenza, Vincenzo A Gennarino, Chiara Di Malta, Francesca Donaudy, Valerio Embrione, Roman S Polishchuk, et al. A gene network regulating lysosomal biogenesis and function. *Science*, 325(5939):473–477, 2009. doi: 10.1126/science.1174447.

[15] Carmine Settembre, Chiara Di Malta, Valentina A Polito, Moises Garcia Arencibia, Francesco Vetrini, Serkan Erdin, Serpil U Erdin, Tuong Huynh, Diego Medina, Pasquale Colella, et al. TFEB links autophagy to lysosomal biogenesis. *Science*, 332(6036):1429–1433, 2011. doi: 10.1126/science.1204592.

[16] Harini Iyer, Kimberle Shen, Ana M Meireles, and William S Talbot. A lysosomal regulatory circuit essential for the development and function of microglia. *Science Advances*, 8(35):eabp8321, 2022. doi: 10.1126/sciadv.abp8321.

[17] Adam Frankish, Mark Diekhans, Irwin Jungreis, Julien Lagarde, Jane E Loveland, Jonathan M

Mudge, Cristina Sisú, James C Wright, Joel Armstrong, If Barnes, et al. GENCODE 2021. *Nucleic Acids Research*, 49(D1):D916–D923, 2021. doi: 10.1093/nar/gkaa1087.

[18] Ari E Friedland, Reshica Baral, Pankhuri Singhal, Katherine Loveluck, Shen Shen, Minerva Sanchez, Eugenio Marco, Gregory M Gotta, Morgan L Maeder, Edward M Kennedy, Anand V R Kornepati, Alexander Sousa, McKensie A Collins, Hari Jayaram, Bryan R Cullen, and David Bumcrot. Characterization of *Staphylococcus aureus* Cas9: a smaller Cas9 for all-in-one adeno-associated virus delivery and paired nickase applications. *Genome Biology*, 16:257, 2015. doi: 10.1186/s13059-015-0817-8.

[19] Yue Zhang, Jiaying Li, Yongfang Zhao, Yichen Huang, Ziyu Shi, Hailian Wang, Hui Cao, et al. Arresting the bad seed: HDAC3 regulates proliferation of different microglia after ischemic stroke. *Science Advances*, 10(10):eade6900, 2024. doi: 10.1126/sciadv.ade6900. ATAC-seq of FACS-sorted microglia from ischemic stroke model (tMCAO). GEO: GSE220041.

[20] Jason D Buenrostro, Paul G Giresi, Lisa C Zaba, Howard Y Chang, and William J Greenleaf. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nature Methods*, 10(12):1213–1218, 2013. doi: 10.1038/nmeth.2688.

[21] M Ryan Corces, Alexandro E Trevino, Emily G Hamilton, Peyton G Greenside, Nicholas A Sinnott-Armstrong, Sam Vesuna, Ansuman T Satpathy, Adam J Rubin, Kathleen S Montine, Beijing Wu, et al. An improved ATAC-seq protocol reduces background and enables interrogation of frozen tissues. *Nature Methods*, 14(10):959–962, 2017. doi: 10.1038/nmeth.4396.

[22] Felix Krueger. Trim Galore, 2015. Software available at https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/.

[23] Ben Langmead and Steven L Salzberg. Fast gapped-read alignment with bowtie 2. *Nature Methods*, 9:357–359, 2012. doi: 10.1038/nmeth.1923.

[24] Heng Li, Bob Handsaker, Alec Wysoker, Tim Fennell, Jue Ruan, Nils Homer, Gabor Marth, Goncalo Abecasis, and Richard Durbin. The sequence alignment/map format and SAMtools. *Bioinformatics*, 25(16):2078–2079, 2009. doi: 10.1093/bioinformatics/btp352.

- [25] Aaron R Quinlan and Ira M Hall. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics*, 26(6):841–842, 2010. doi: 10.1093/bioinformatics/btq033.
- [26] Haley M Amemiya, Anshul Kundaje, and Alan P Boyle. The ENCODE blacklist: identification of problematic regions of the genome. *Scientific Reports*, 9:9354, 2019. doi: 10.1038/s41598-019-45839-z.
- [27] John M Gaspar. Genrich: detecting sites of genomic enrichment, 2019. Software available at <https://github.com/jsh58/Genrich>.
- [28] Yong Zhang, Tao Liu, Clifford A Meyer, Jérôme Eeckhoutte, David S Johnson, Bradley E Bernstein, Chad Nusbaum, Richard M Myers, Myles Brown, Wei Li, and X Shirley Liu. Model-based analysis of ChIP-Seq (MACS). *Genome Biology*, 9(9):R137, 2008. doi: 10.1186/gb-2008-9-9-r137.
- [29] Ben Langmead, Cole Trapnell, Mihai Pop, and Steven L Salzberg. Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biology*, 10(3):R25, 2009. doi: 10.1186/gb-2009-10-3-r25.
- [30] Felix Mölder, Kim Philipp Jablonski, Brice Letcher, Michael B Hall, Christopher H Tomkins-Tinch, Vanessa Sochat, Jan Forster, Soohyun Lee, Sven O Twardziok, Alexander Kanitz, et al. Sustainable data analysis with Snakemake. *F1000Research*, 10:33, 2021. doi: 10.12688/f1000research.29032.2.

Figure legends

Figure 1. Corrected analysis architecture and scope. Reference/TSS selection and five targeting classes are integrated with uniformly reprocessed public ATAC-seq. The primary call requires the complete protospacer plus PAM inside a primary peak.

Figure 2. Sequence coverage and genome-wide guide-site support. (A) Genome-wide and (B) frozen-panel promoters with at least one passing candidate. (C) Primary genome-wide calls by context and targeting class. (D) Candidate-count distributions.

Figure 3. Sequence availability versus primary guide-site support. (A) Un1Cas12f1 sequence coverage and complete-site support in the frozen panel. (B) Panel summaries for five targeting classes. (C) Gene-level Un1Cas12f1 matrix. Between-context differences are descriptive.

Figure 4. Descriptive panel patterns and TSS robustness. (A) Cross-context support patterns. (B) *Tfeb/Tfe3* primary calls and promoter-overlap context. (C) TSS offsets. (D) Call stability across TSS, depth, and caller variants.

Figure 5. Evidence-aware candidate prioritization. (A) Experimental-prioritization workflow. (B) Top Un1Cas12f1 *Tfe3* candidates with supporting signal and preliminary PAM-aware off-target counts.

Figure 6. Canonical-TSS-centered ATAC signal and guide evidence at *Tfeb* and *Tfe3*. CPM-normalized signal within ± 2.5 kb, primary peak intervals, promoter windows, and supported candidate positions. All tracks use the corrected canonical TSS and primary peak rules. The six contexts share a y-axis scale within each gene column; the two gene columns are scaled independently.

Supplementary Figure S1. Per-run ATAC-seq quality control. Alignment, FRiP, TSS enrichment, and usable units; technical runs are visually distinguished from biological replicates.

Supplementary Figure S2. Genome-wide candidate distributions and Un1Cas12f1 primary calls.

Supplementary Figure S3. Operational-definition sensitivity. Complete containment, partial guide overlap, promoter-midpoint coverage, and any promoter overlap are shown separately.

Supplementary Figure S4. TSS, depth, and peak-caller sensitivity of frozen-panel calls.

492 **Supplementary Figure S5. Functional-category associations within the frozen 55-**
493 **gene universe.**

494 **Supplementary Figure S6. Matched-depth Genrich/MACS3 concordance and de-**
495 **terministic sampling fractions.**

496 **Supplementary Figure S7. Promoter-matched null specification and results.**

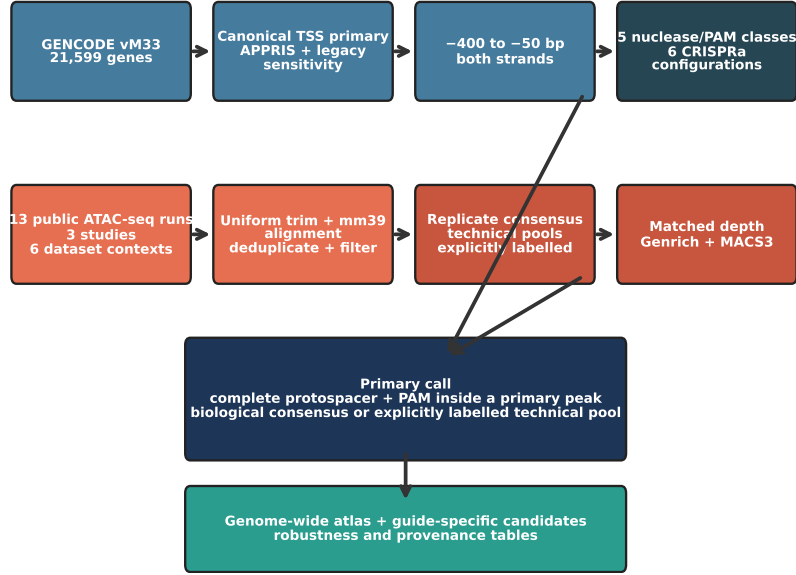
Table 1: Targeting classes and represented modelled CRISPRa configurations. HEAL and SminiCRa share one targeting class; the dNme2Cas9 activator is proposed rather than experimentally validated.

Targeting class	PAM	Spacer	Represented configuration(s)
Un1Cas12f1	TTTR, 5-prime	20 nt	HEAL; SminiCRa
SaCas9	NNGRRT, 3-prime	21 nt	SaCas9 CRISPRa
SpCas9	NGG, 3-prime	20 nt	SpCas9 CRISPRa
CjCas9	NNNVRYM, 3-prime	22 nt	MiniCAFE
Nme2Cas9	NNNNCC, 3-prime	24 nt	proposed dNme2Cas9 activator

Table 2: Public ATAC-seq inputs. “Context” is descriptive and does not imply a balanced cross-study state comparison.

Study	Context	Runs	Layout	Replication used
Gosselin 2017	ex vivo homeostatic	2	single-end	biological
Zhang X/Kracht 2022	PBS/PBS	2	paired-end	technical
Zhang X/Kracht 2022	PBS/LPS	2	paired-end	technical
Zhang X/Kracht 2022	LPS/LPS	2	paired-end	technical
Zhang 2024	sham	2	paired-end	biological
Zhang 2024	stroke	3	paired-end	biological

A Corrected analysis architecture



B Scope

- 21,599** protein-coding promoters
- 5** nuclease/PAM classes
- 6** CRISPRa configurations
- 6** surveyed dataset contexts
- 13** raw sequencing runs
- 55** locked therapeutic genes
- 3** TSS definitions
- 2** matched-depth peak callers

Figure 1

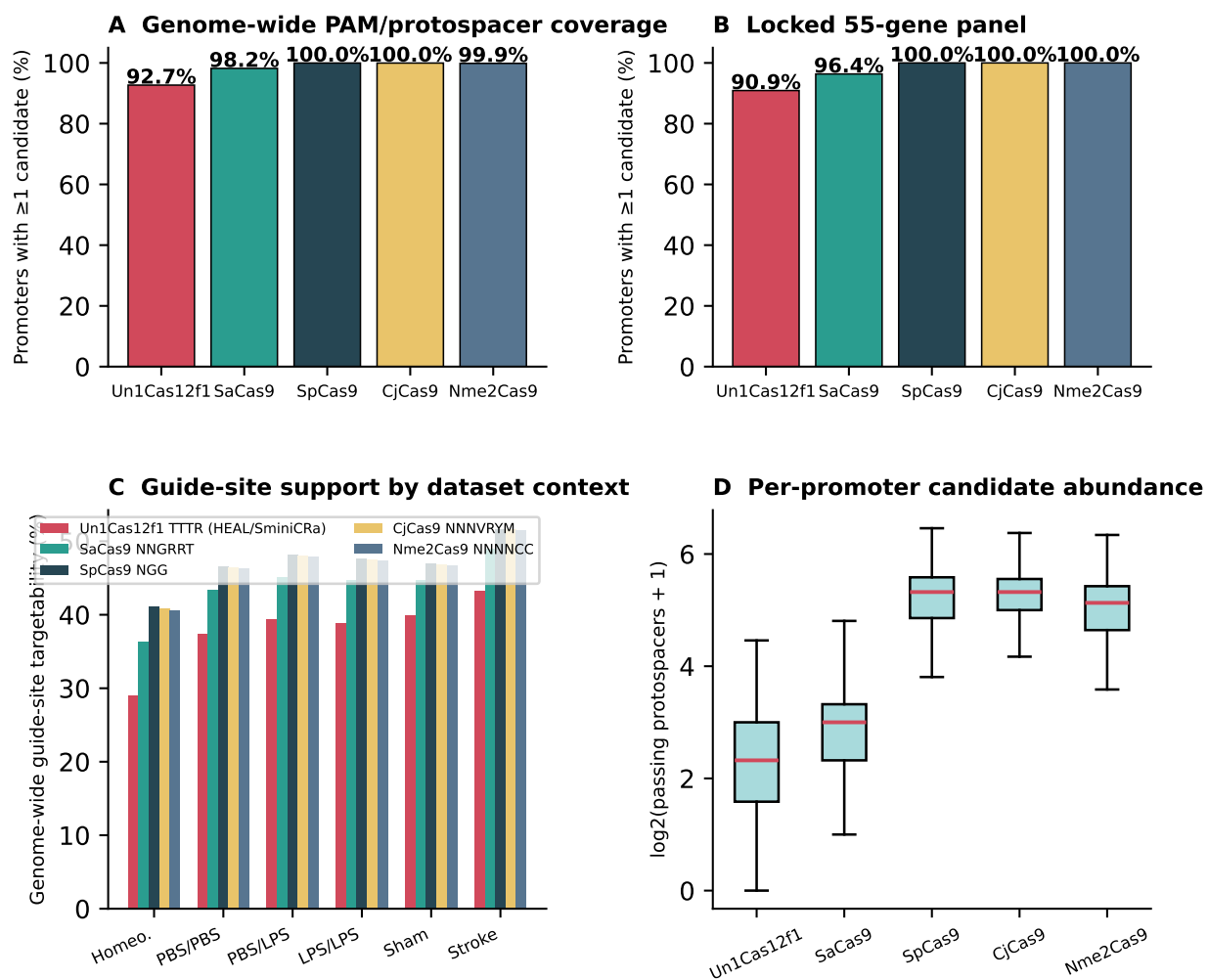


Figure 2

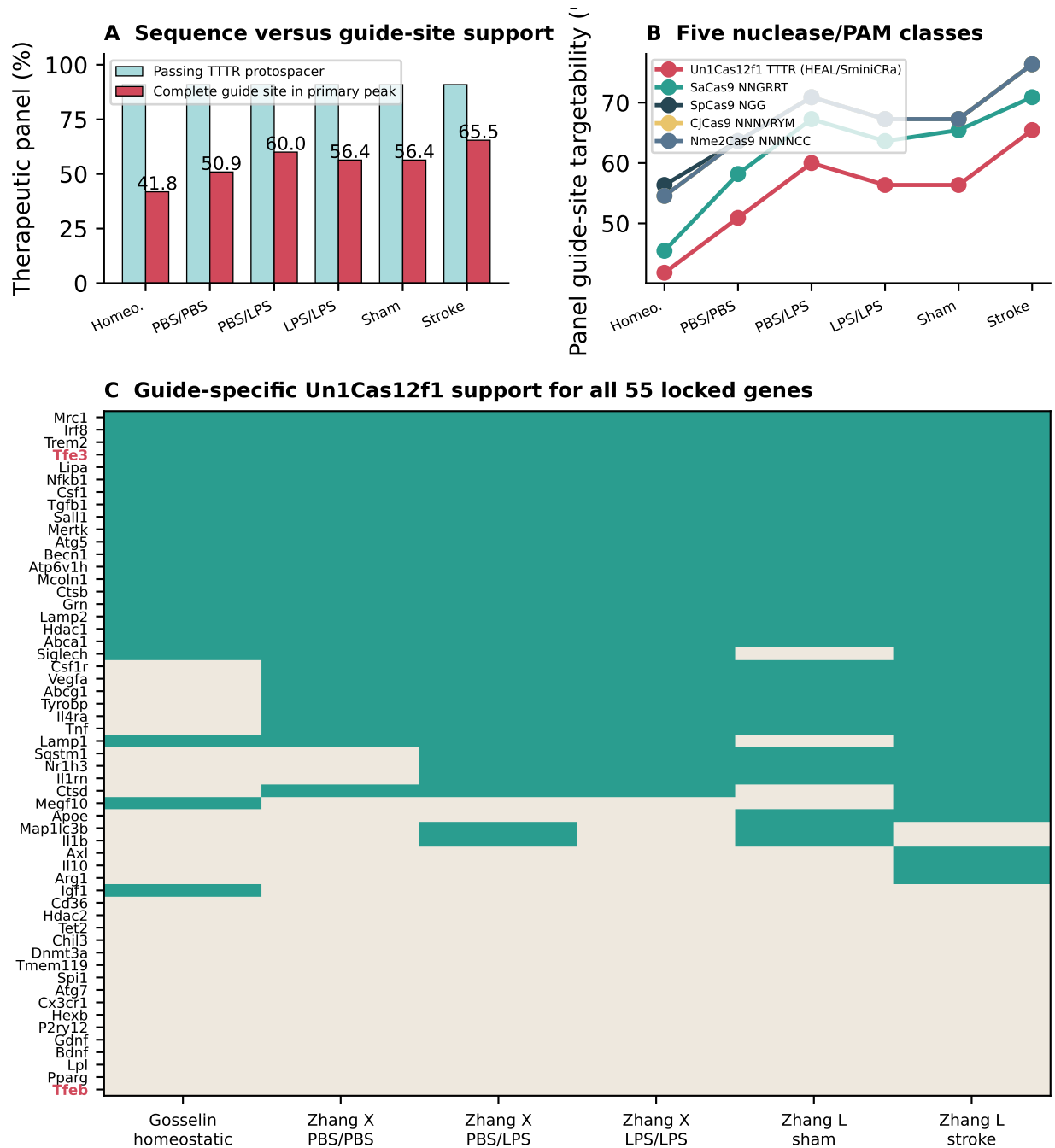


Figure 3

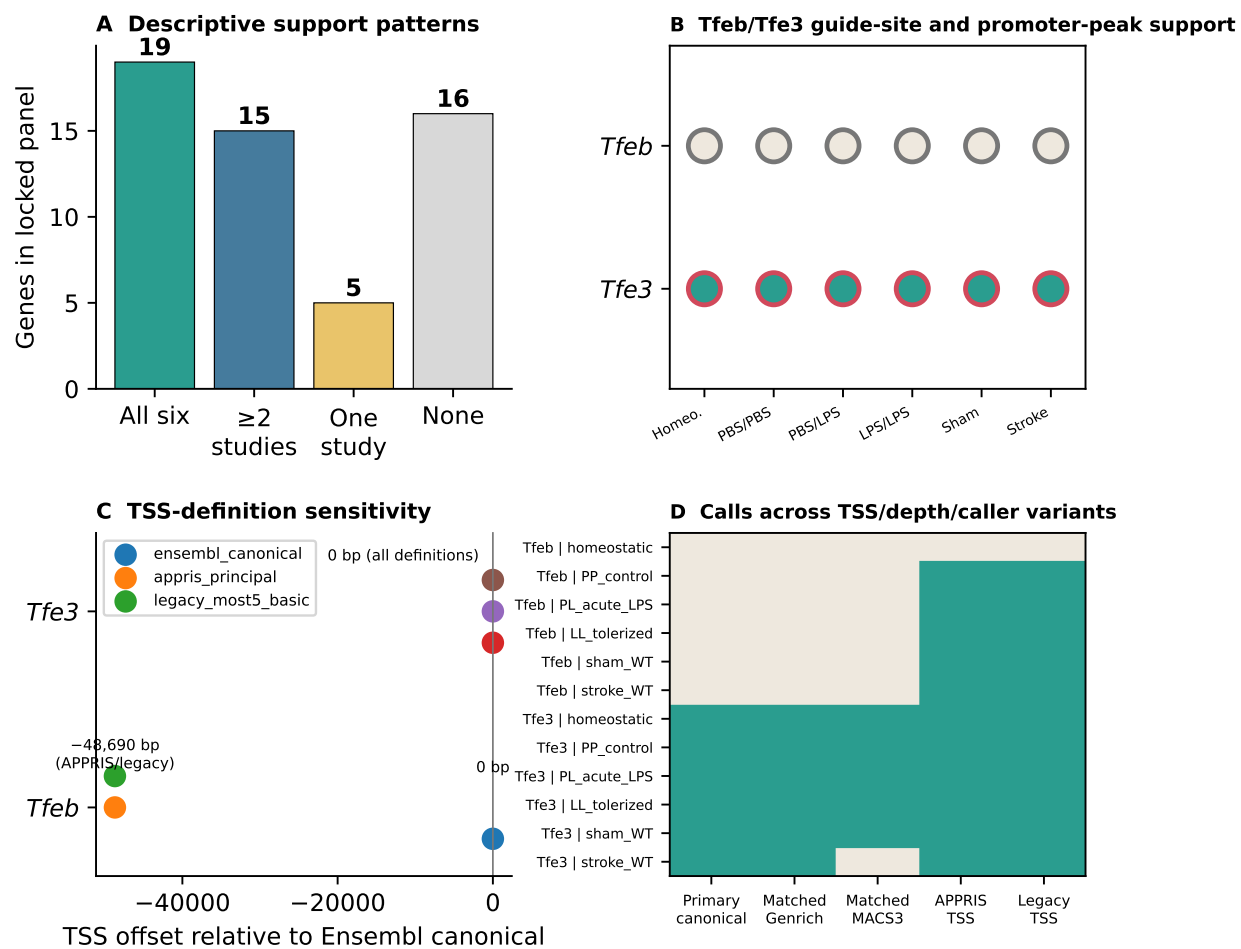
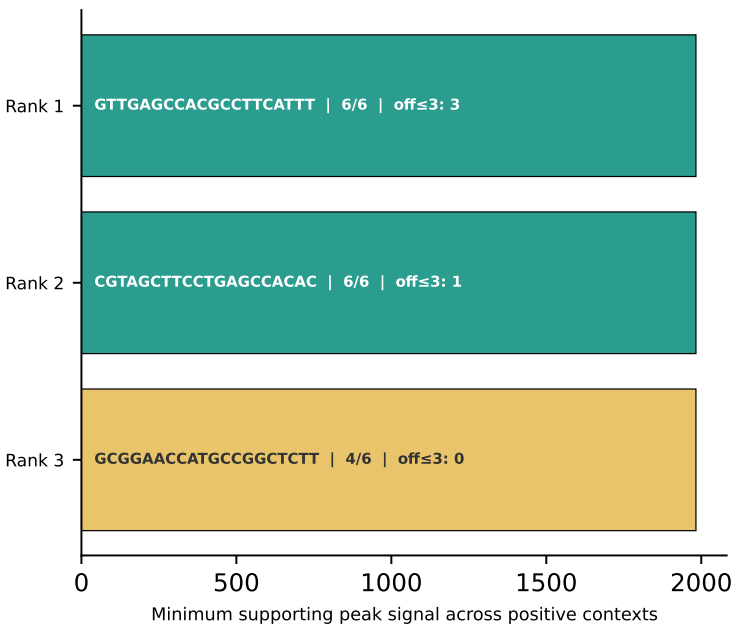


Figure 4

A Evidence-aware prioritization

- 1 Choose delivery architecture**
Size and payload constraints
- 2 Choose expressed TSS**
Canonical + cell-state sensitivity
- 3 Require guide-site support**
Complete spacer+PAM in peak
- 4 Inspect robustness**
Replicates, depth, caller, context
- 5 Validate experimentally**
Activation, function and safety

B Tfe3 Un1Cas12f1 candidate protospacers



Predictive ranking; PAM-aware Bowtie screening does not replace purpose-built off-target tools or experimental safety assays.

Figure 5

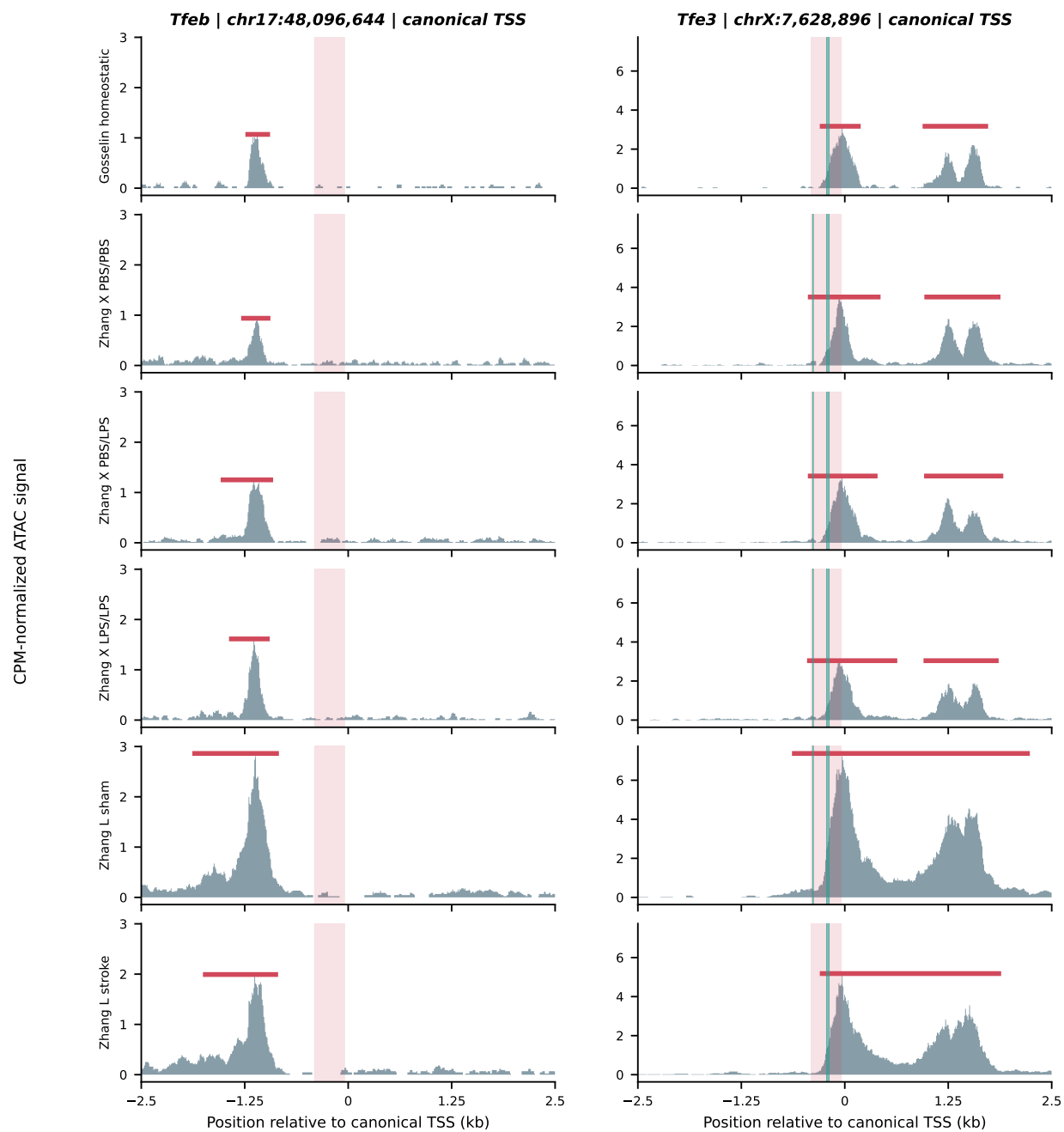


Figure 6